

AIDAN M. MOKALLA

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UNDERGRADUATE EDUCATION

GPA: 3.6/4

- Reed College, Portland, Oregon, USA,2025
- + Spanish minor, sole member of graduating class with double majors: Computer Science & Linguistics,
- + A grades in Real Analysis, Linear Algebra, Deep Learning, Linguistic Field Methods, and Psycholinguistics,
- Bryn Mawr College, Dushanbe, Tajikistan,2024
- + Fellowship funded by international non-profit American Councils to research Persian language varieties,
- La Universidad de Carlos III and Middlebury College, Madrid, Spain,2023
- + Spanish-language courses including Artificial Intelligence, Data Science, and International IP Law.

SELECTED PUBLICATION

- Source of the Storm: Interpreting Weather Models to Detect Cyclogenesis,forthcoming
- + Applying post-superposition statistical interpretation methods to weather prediction models to improve localization of rare climatic events and demonstrate methodological generalizability beyond transformer-based models,
- Evaluating Diffusion Language Models Beyond Benchmarks,CoLM 2026
- + Found that people are more likely to over-trust AI outputs from the recently invented diffusion-based LLM architecture than when evaluating the outputs of standard autoregressive LLMs,
- Interpretable Signals for VLA Model Failure,ICML 2026
- + Discovered representational mechanistic predictors of vision-language-action model task failure,
- Fault- and Network-Agnostic Agreement, with Erica Blum,forthcoming
- + Proved tight running-time bounds for all algorithms achieving network-agnostic Byzantine consensus,
- Language-Agnostic Concept Spaces Found in Large Language Models, with Greg Anderson, thesis, 2025
- + Designed, implemented, and evaluated novel architecture to interpret transformer-based LLMs by leveraging natural-language translation of ground-truth semantics. My approach learns convergent “concept spaces” over LLM activations in parallel, yielding more consistent LLM interpretations than prior work,
- Modulated Signal Filtering: Wouldn't it be Noise?, with Sameer ud Dowla Khan,thesis, 2025
- + Collected over 3,000 participant responses to investigate how listeners use unique cognitive strategies to parse spoken speech versus to parse song lyrics, using both deep and symbolic statistical methods,
- + Demonstrated that we use contextually-differentiated mechanisms to filter noise from the “linguistic signal” by making significantly more accurate predictions of how people interpret unclear speech under this prior.

LEADERSHIP & EXPERIENCE

- Algoverse AI Research, Machine Learning Researcher, Fellow, & MentorSeptember 2025 – present
- + Investigating and publishing methods of explainable AI and climate science,
- Linguistic Diversity Initiative, Lead Volunteer Ambassador,June 2023 – August 2025
- + Leading a team of three translators, I compiled and published dictionaries of terminology used in Portland Public’s science curricula, translated into 6 languages, for households with low English proficiency, already enabling over 800 families to engage in their students’ education,
- + Taught seminars to students and faculty about sociolinguistic research on language use in undergraduate settings,
- + Organized and lead twice-monthly meetings of over 50 volunteers and larger events,
- Reed College Student Body, Senator & Honor Councillor,September 2021 – May 2025
- + Peer-elected to serve in faculty committees including the Committee on Academic Policies and Planning and Committee on Computer Policy and Planning, disburse 6-figure Student Body Fund and other resources to college organizations, and interview and hire all student applicants for on-campus roles,
- + Mediated and resolved conflicts between students, professors, and other members of the college community,
- Stoel Rives, LLC, Intern Practise Assistant,May – August 2022
- + Implemented compliant automation of docketing, court appearances, and other events from firm-wide databases,
- + Maintained communication on behalf of attorneys while drafting and proofreading attorney correspondences.

SKILLS & ACCOLADES

- Computational: Python, various data and NLP pipelines, PyTorch, NumPy, JS, Next.js, C++, PHP, SQL, Swift,
- Awards: Presidential Commendation for Academic Excellence, Reed College Music Scholarship, National Diversity in STEM Grant, Department of State Critical Languages Scholarship,
- Certifications: Federally eligible for non-competitive civil service, licensed in Conflict Mediation & Resolution, CPR,
- Languages: Fluent Spanish, English, intermediate Farsi, Tajik, French, basic Portuguese, Dari, Chinese (Mandarin).